

Bluestock Fintech

Mutual Fund Analytics Platform

Final Capstone Project Report

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Domain: Mutual Fund / Fintech | Duration: 7 Days

1. Executive Summary

This project implements an end-to-end Mutual Fund Analytics Platform for Bluestock Fintech, covering data ingestion, cleaning, database design, exploratory analysis, performance and risk metrics, an interactive Power BI dashboard, and advanced investor analytics. The platform tracks 40 real mutual fund schemes across 10 AMCs, with NAV history spanning January 2022 to May 2026, alongside industry-wide AUM, SIP and folio growth data sourced from AMFI India and mfapi.in.

Metric	Value
Total schemes tracked	40
Fund houses (AMCs)	10
NAV history records	46,000
Investor transactions	32,778
Unique investors	5,000
Industry AUM (Dec 2025)	Rs. 81 lakh crore
Monthly SIP inflow (Dec 2025)	Rs. 31,002 crore
Total MF folios (Dec 2025)	26.12 crore

2. Data Sources & Datasets

All data used in this project is sourced from publicly available AMFI India data, the mfapi.in REST API, and NSE/BSE benchmark indices. Ten datasets were provided covering fund master details, NAV history, AUM by fund house, monthly SIP inflows, category-wise inflows, industry folio counts, scheme performance metrics, investor transactions, portfolio holdings, and benchmark index prices.

2.1 Live API Integration

In addition to the provided CSV files, live NAV data was fetched directly from the mfapi.in REST API (no authentication required) for 6 key large-cap schemes including HDFC Top 100, SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip and Kotak Bluechip, demonstrating real-time data ingestion capability.

3. ETL Pipeline

The ETL pipeline follows the standard Extract, Transform, Load pattern implemented in Python using Pandas and SQLAlchemy.

3.1 Extract

`data_ingestion.py` loads all 10 provided CSV datasets, prints shape, data types and head for each, and flags anomalies such as missing values, duplicates and date columns stored as text.

`live_nav_fetch.py` calls the `mfapi.in` API to fetch live NAV history for 6 schemes and saves them as raw CSVs.

3.2 Transform

`data_cleaning.py` converts all date columns to proper datetime objects, sorts NAV data by fund and date, forward-fills missing NAV values for weekends and holidays, removes duplicate rows, validates that NAV values are positive, standardises transaction types to SIP/Lumpsum/Redemption, validates transaction amounts and expense ratios against SEBI ranges (0.1% to 2.5%), and computes daily return percentages for every fund.

3.3 Load

`database_setup.py` creates a SQLite database (`bluestock_mf.db`) using a star schema with one dimension table (`dim_fund`) and six fact tables (`fact_nav`, `fact_transactions`, `fact_performance`, `fact_portfolio`, `fact_aum`, `fact_sip_industry`). All cleaned CSVs are loaded via SQLAlchemy, totalling 79,318 rows across the database, with indexes on `amfi_code` and date columns for query performance.

4. Exploratory Data Analysis Findings

Comprehensive EDA was performed across NAV history, AUM growth, SIP inflows, investor demographics, geography, and portfolio composition. Nine publication-quality charts were produced covering all required categories.

- 1.** All 40 equity funds showed consistent upward NAV trends from 2022 to 2026, with a strong 2023 rally and minor corrections in early 2024.
- 2.** SBI Mutual Fund is the largest AMC with Rs.12.5 lakh crore AUM as of Dec 2025, roughly 25% larger than the second-largest, ICICI Prudential.
- 3.** Monthly SIP inflows grew 169% from Rs.11,517 crore (Jan 2022) to Rs.31,002 crore (Dec 2025), reflecting India's deepening retail equity culture.
- 4.** Large Cap funds receive the highest and most consistent category inflows, followed by Flexi Cap and Mid Cap.
- 5.** The 26-35 age group has the largest number of SIP investors (1,937), while the 56+ group invests the highest average amount per SIP (Rs.11,575).
- 6.** Punjab leads total investment by state, closely followed by Tamil Nadu and Madhya Pradesh, across 12 states covered.
- 7.** T30 cities contribute the majority of total investment value, though B30 cities represent a growing share - a key focus area for AMC expansion.
- 8.** Total MF folios nearly doubled from 13.26 crore (Jan 2022) to 26.12 crore (Dec 2025), a 97% increase in investor base over 4 years.
- 9.** Equity funds within the same sub-category show NAV return correlations of 0.85 to 0.95, indicating limited diversification benefit within a category.
- 10.** Banking and Financial Services is the dominant sector across equity fund portfolios, typically representing 25-30% of total portfolio weight.

4.1 NAV Trends - All Equity Funds

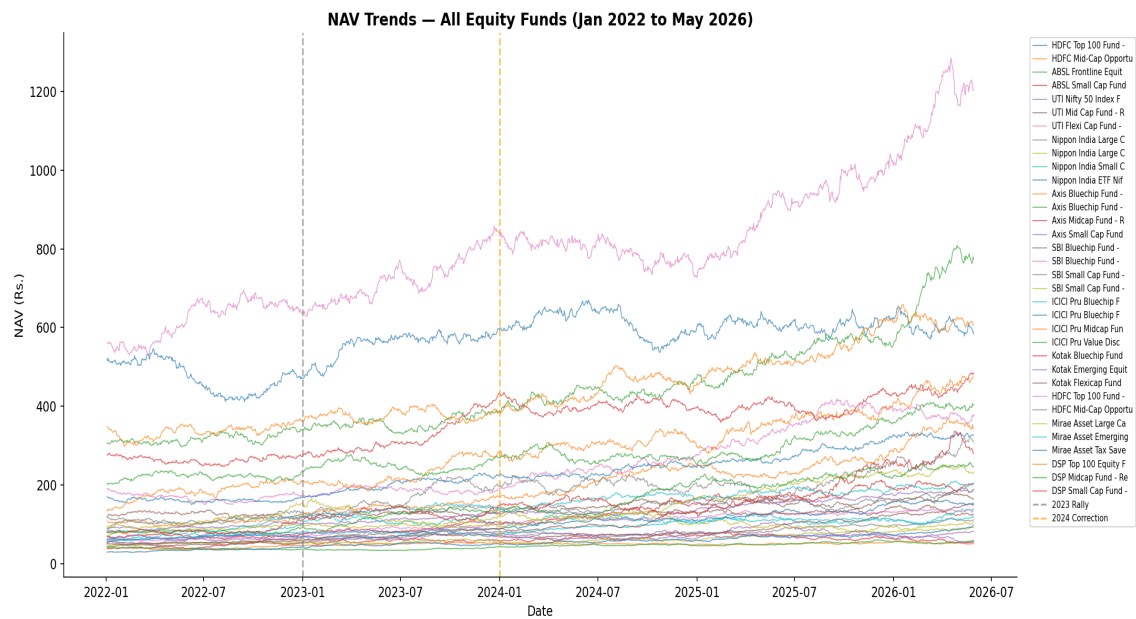


Figure 1: Daily NAV for all equity schemes, Jan 2022 - May 2026

4.2 AUM Growth by Fund House

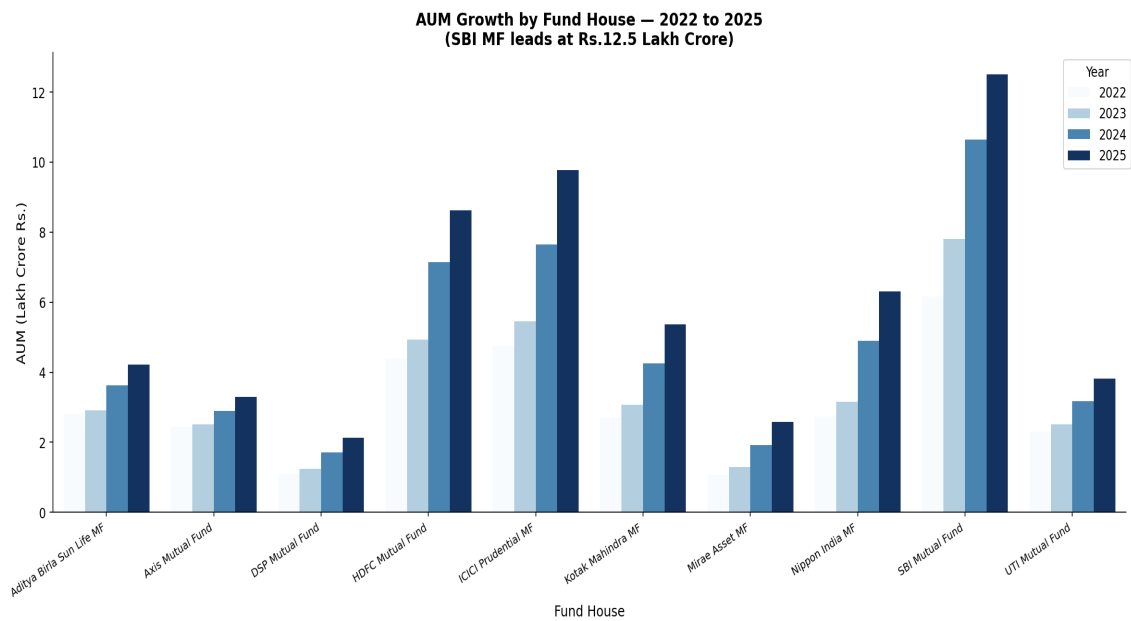


Figure 2: AUM by fund house, 2022-2025

4.3 SIP Inflow Trend

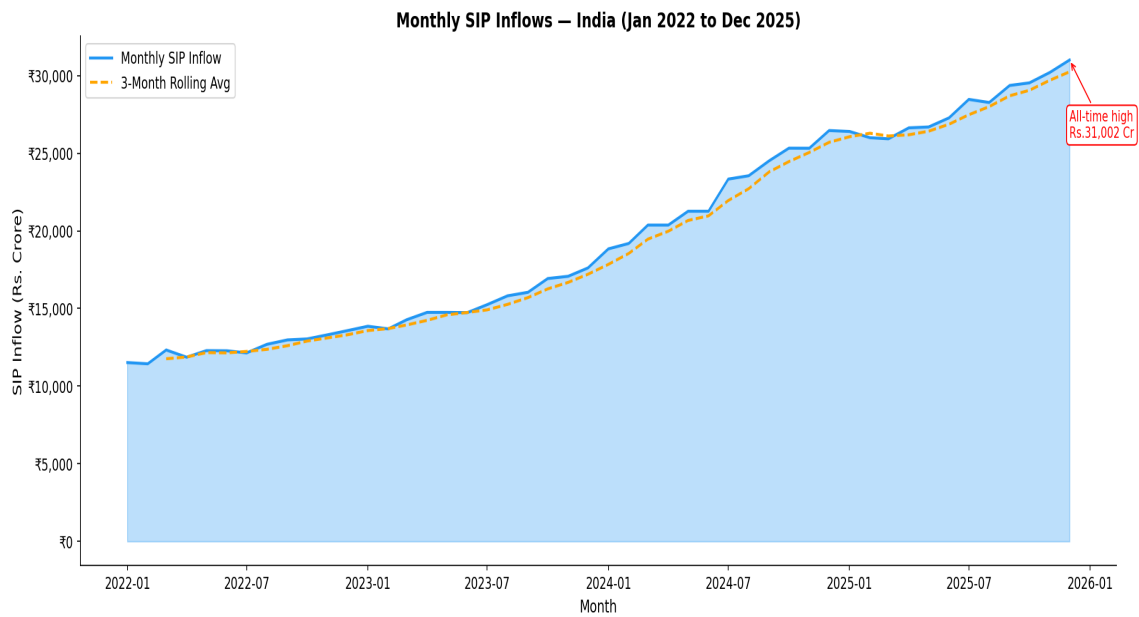


Figure 3: Monthly SIP inflows, marking the Rs.31,002 crore all-time high

4.4 Category Inflow Heatmap

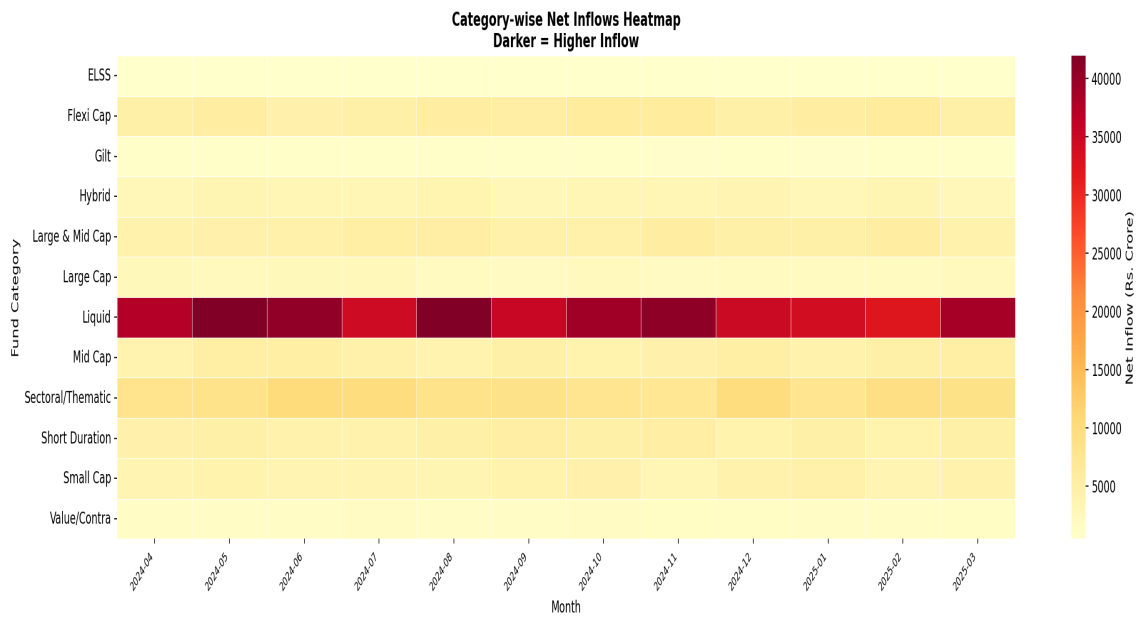


Figure 4: Category-wise net inflows by month

4.5 Investor Demographics

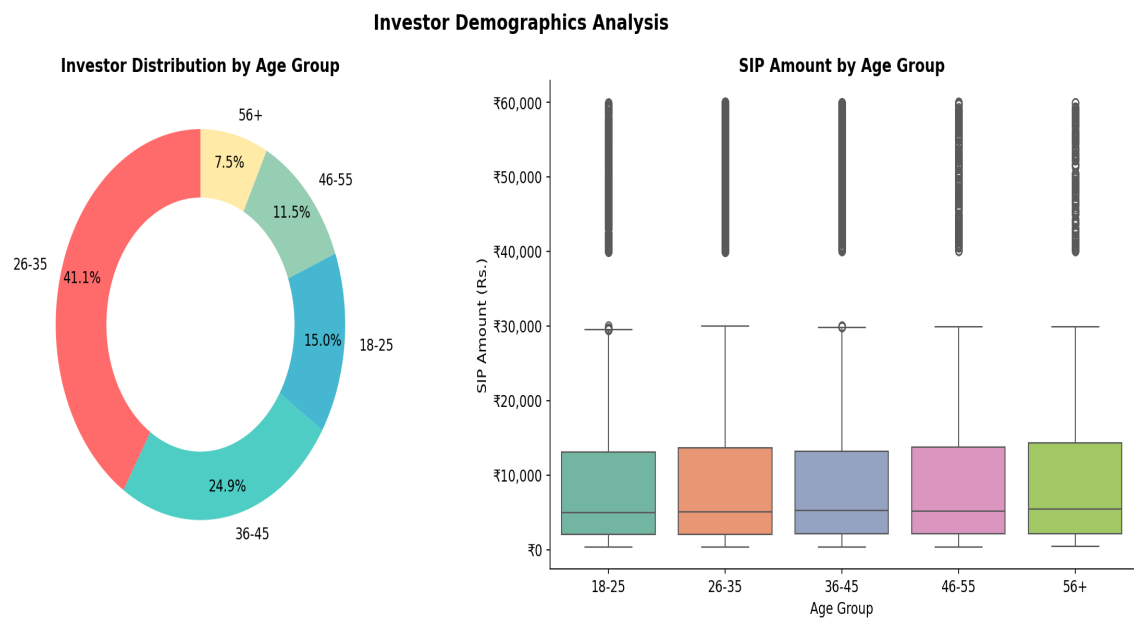


Figure 5: Age group distribution and SIP amount by age group

4.6 Geographic Distribution

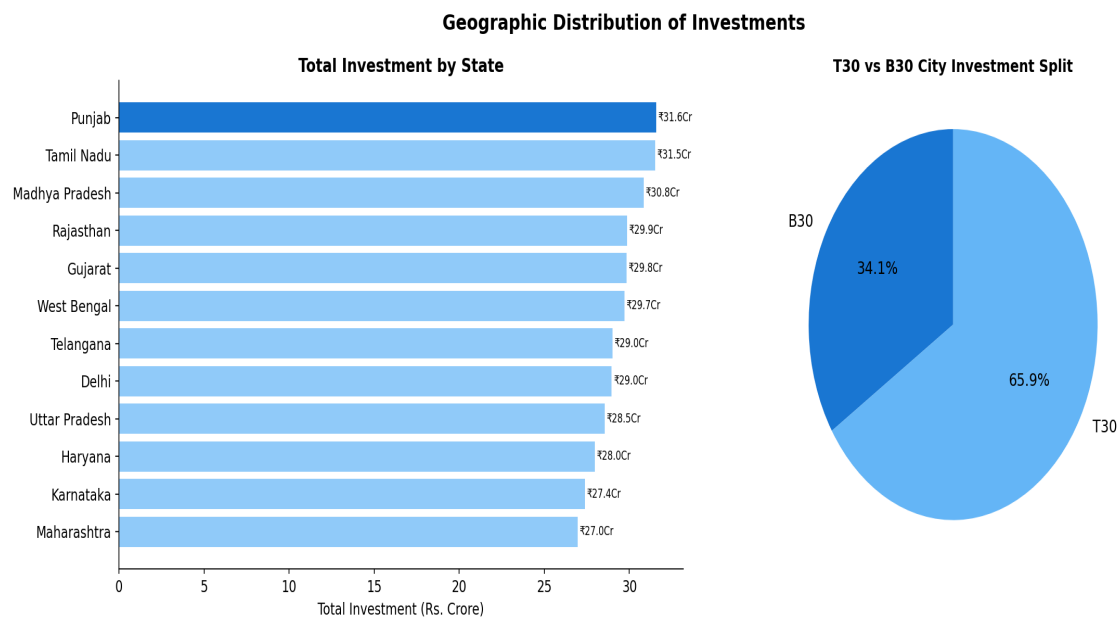


Figure 6: Investment by state and T30 vs B30 split

4.7 Folio Count Growth

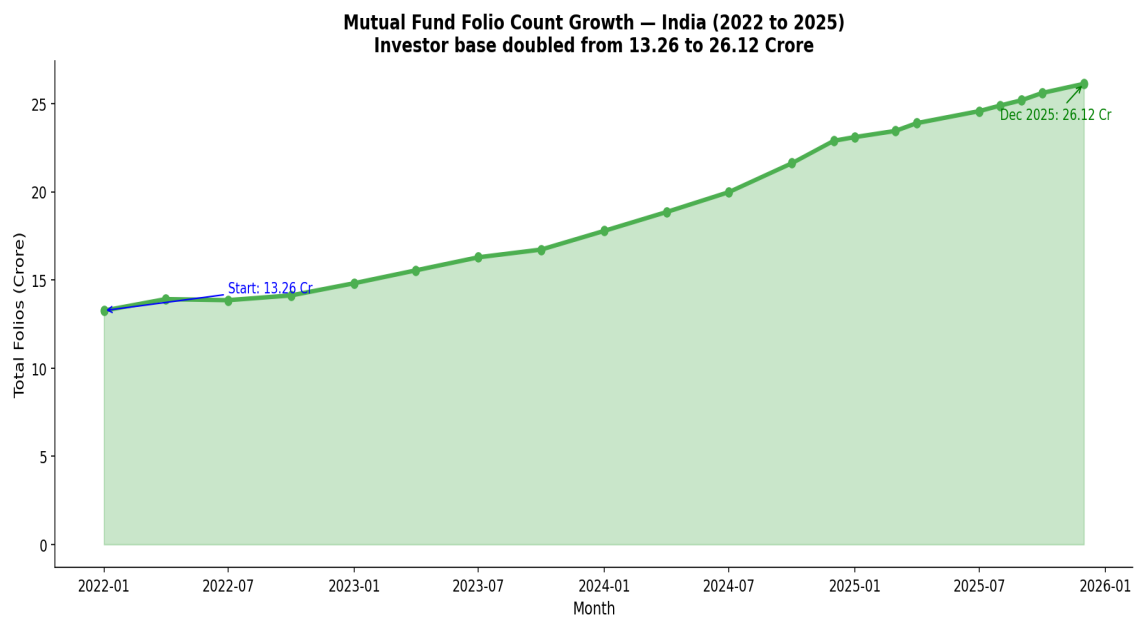


Figure 7: Total MF folios growth, 2022-2025

4.8 NAV Return Correlation Matrix

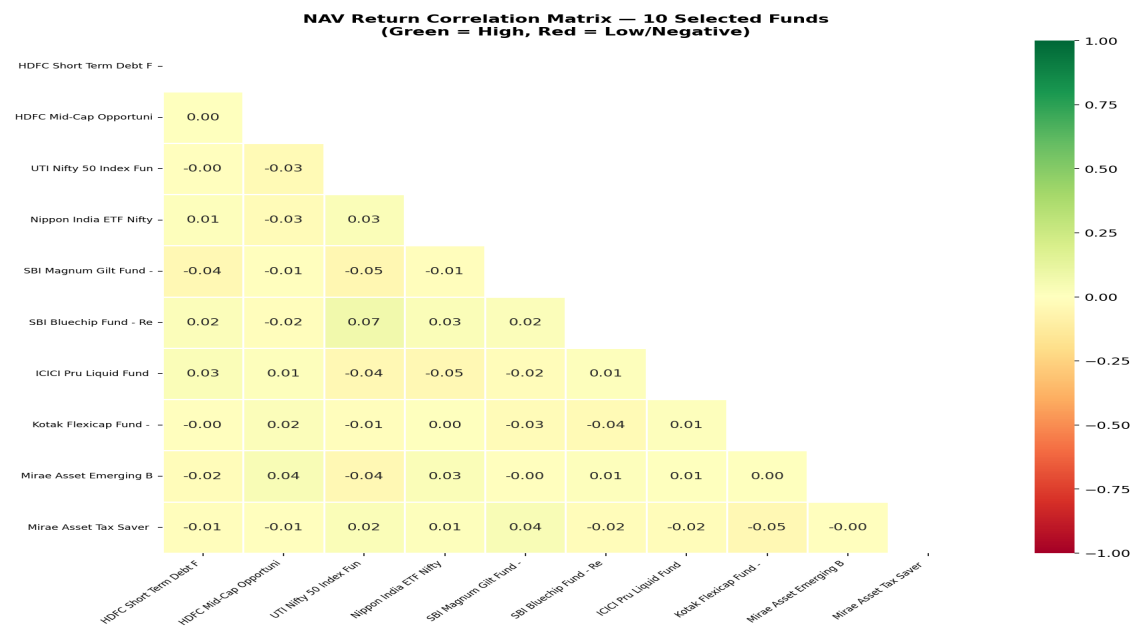


Figure 8: Correlation of daily returns across 10 selected funds

4.9 Sector Allocation

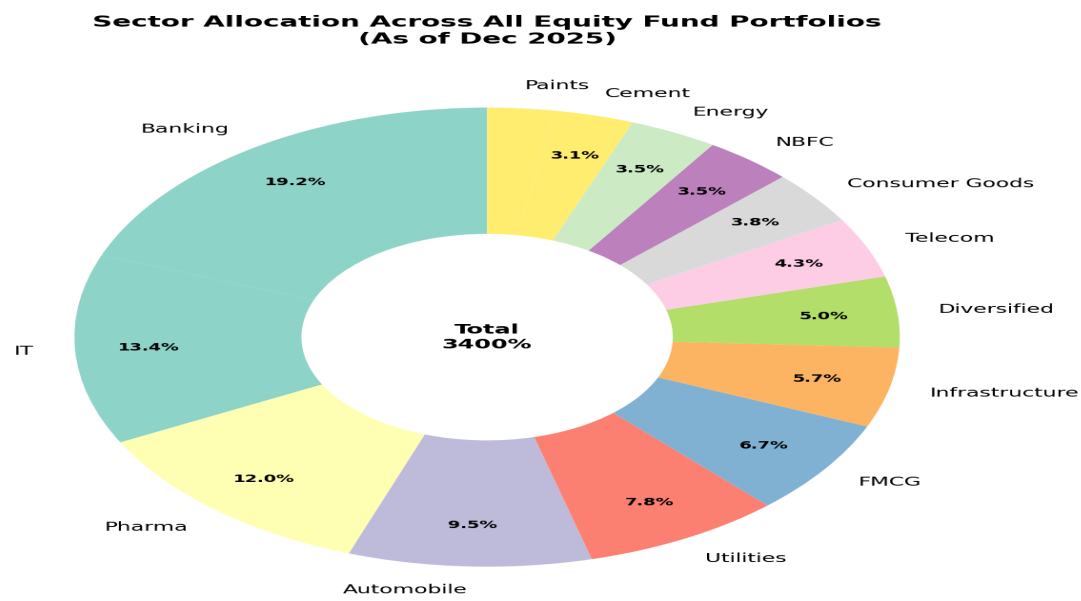


Figure 9: Sector weights across all equity fund portfolios

5. Performance & Risk Analysis

Key performance and risk metrics were computed from NAV history for all 40 funds, including annualised returns, CAGR, Sharpe ratio, Sortino ratio, Alpha, Beta, and Maximum Drawdown. A composite Fund Scorecard ranks all funds on a 0-100 scale combining these metrics with weighted importance.

5.1 Top 10 Funds by Composite Score

Rank	Scheme	Score	3yr Ret%	Sharpe	Alpha
1	Kotak Flexicap Fund - Regular - Gro	76.0	15.65	1.31	1.85
2	SBI Small Cap Fund - Regular Plan -	73.9	23.39	0.95	1.23
3	ICICI Pru Midcap Fund - Regular - G	69.0	18.08	1.18	0.89
4	ABSL Small Cap Fund - Regular - Gro	68.8	22.38	0.16	1.84
5	SBI Small Cap Fund - Direct Plan -	67.4	23.14	-0.06	1.13
6	Axis Midcap Fund - Regular - Growth	66.8	15.18	1.00	1.42
7	Mirae Asset Large Cap Fund - Regula	66.2	14.81	1.45	1.62
8	HDFC Mid-Cap Opportunities Fund - R	66.0	16.58	1.09	0.95
9	Kotak Emerging Equity Fund - Regula	63.1	18.23	0.08	1.91
10	DSP Small Cap Fund - Regular - Grow	60.9	20.08	0.95	0.69

5.2 Fund Scorecard Visualisation

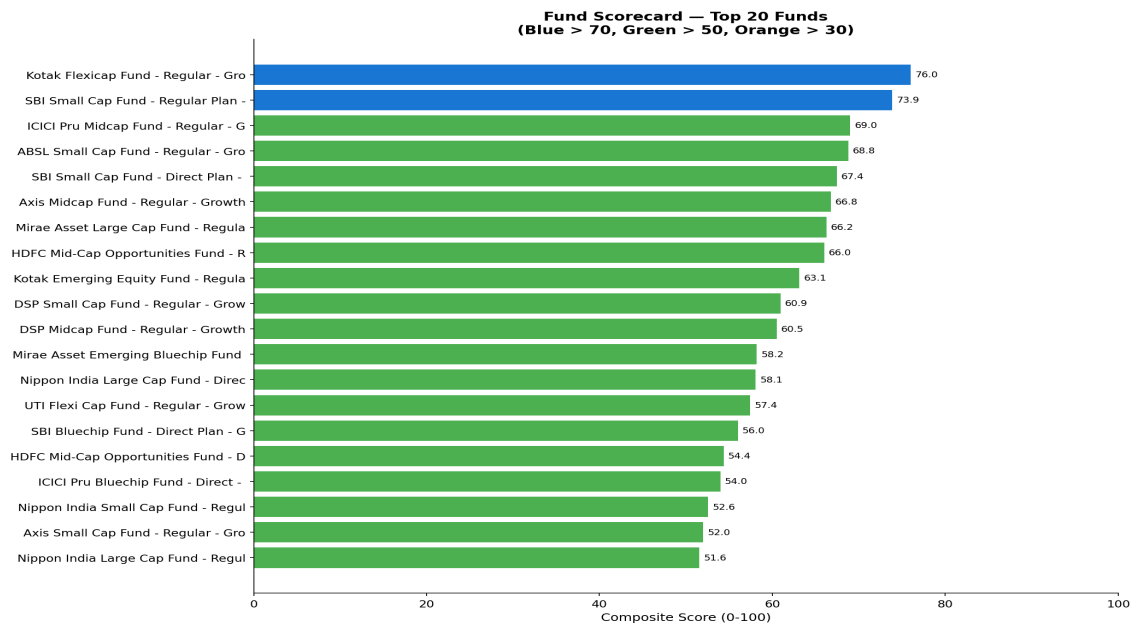


Figure 10: Composite scorecard - top 20 funds

5.3 Benchmark Comparison

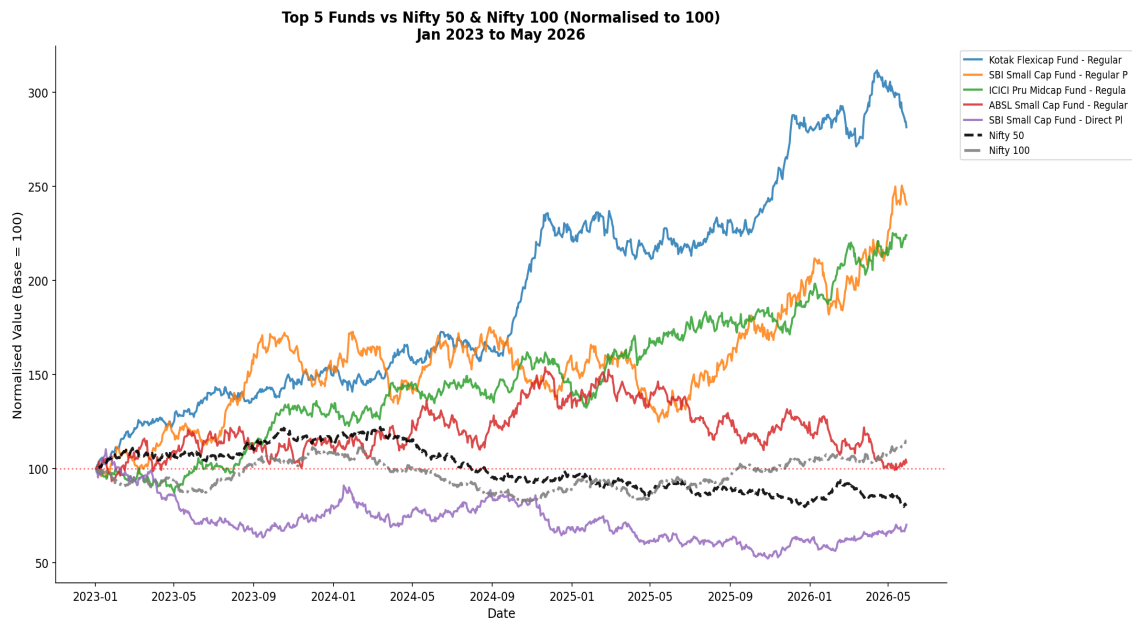


Figure 11: Top 5 funds vs Nifty 50 and Nifty 100 (normalised to 100)

5.4 Risk Analysis - VaR and CVaR

Historical Value at Risk (95%) and Conditional VaR were computed for all funds. Small Cap funds (SBI, Axis, ABSL, Nippon, DSP) show the highest tail risk with worst single-day losses exceeding 4.5%, while Liquid funds (ICICI Pru Liquid, Kotak Liquid) are the safest with worst-day losses under 0.1%.

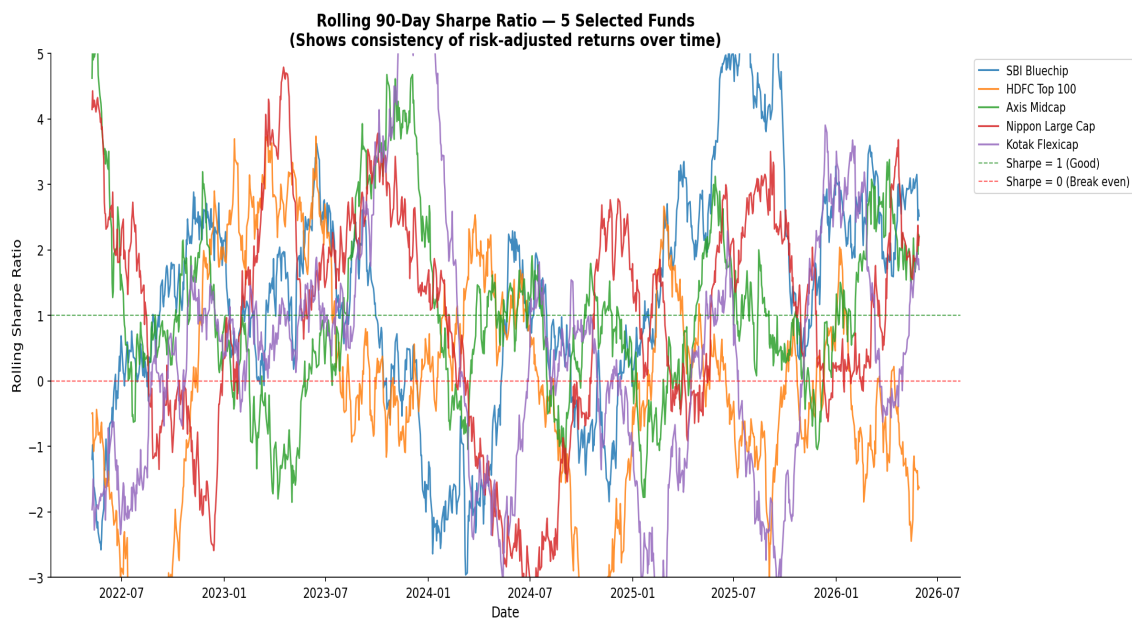


Figure 12: Rolling 90-day Sharpe ratio for 5 selected funds

5.5 Sector Concentration (HHI)

Herfindahl-Hirschman Index was computed for all equity fund portfolios. All 34 funds with portfolio data scored below the High Concentration threshold of 2500, with Axis Bluechip the most concentrated (HHI 2064, 38% in Infosys) and SBI Small Cap Regular the most diversified (HHI 1073).

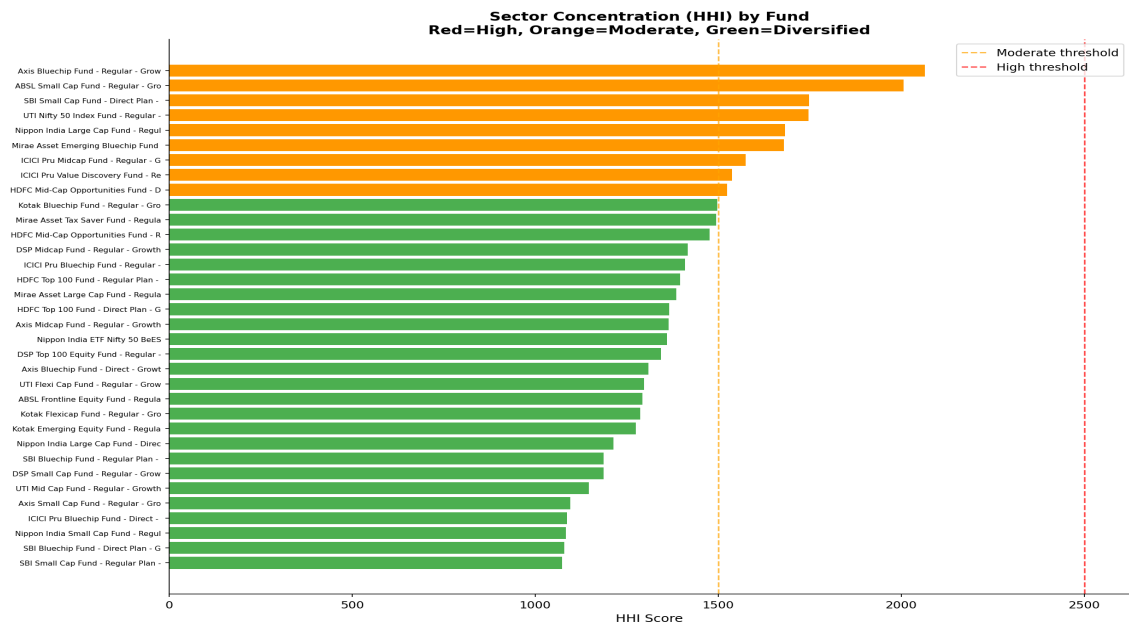
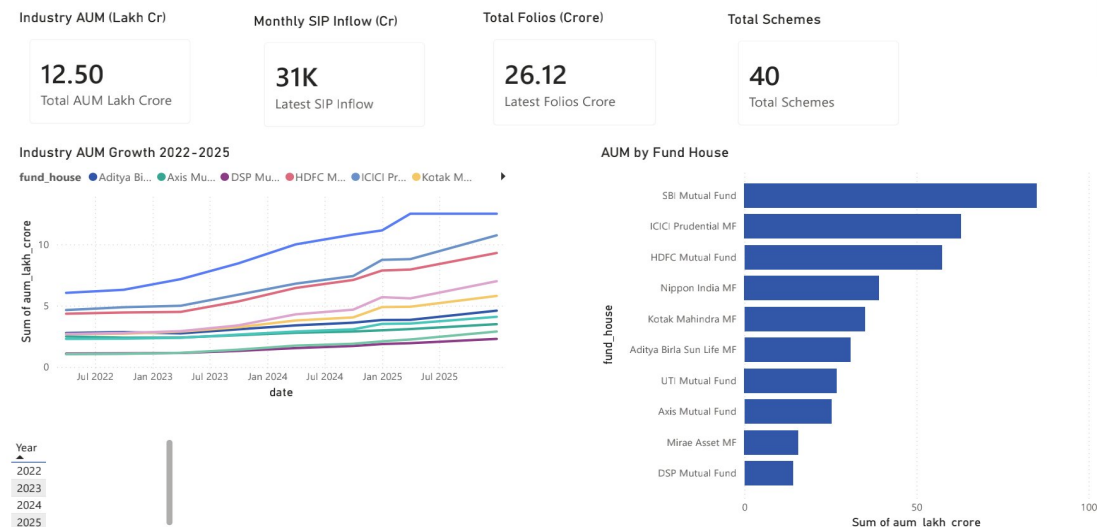


Figure 13: Sector concentration (HHI) by fund

6. Interactive Dashboard

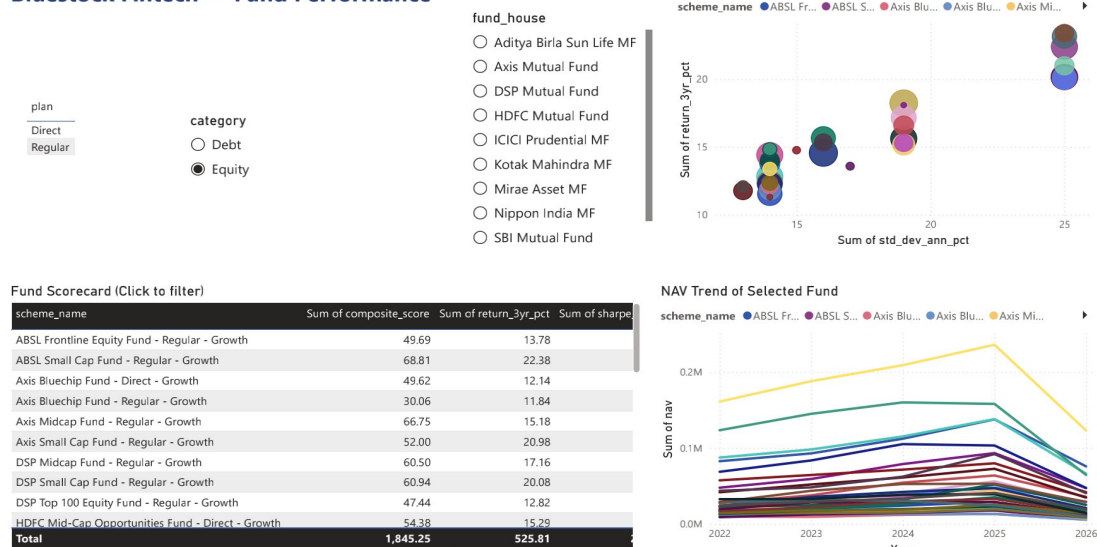
A 4-page interactive Power BI dashboard was built covering Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends. Every page includes at least 2 interactive slicers for filtering by fund house, category, state, age group, and time period.

Bluestock Fintech — Industry Overview



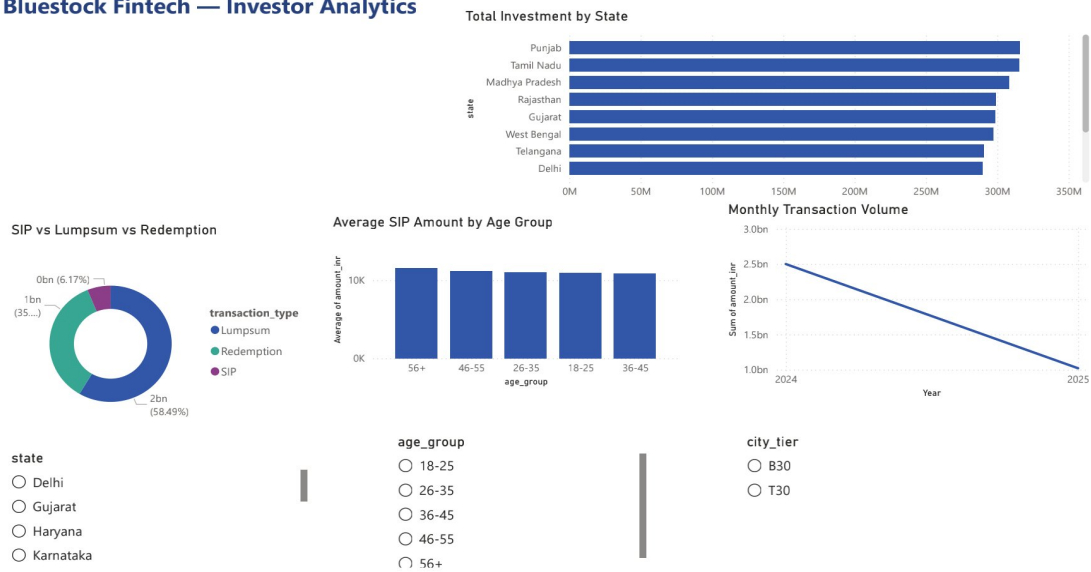
Page 1: Industry Overview - KPI cards for Total AUM, SIP Inflows, Folios, and Schemes; AUM trend and fund house comparison.

Bluestock Fintech — Fund Performance



Page 2: Fund Performance - Return vs Risk scatter, sortable scorecard, NAV trend with Fund House/Category/Plan slicers.

Bluestock Fintech — Investor Analytics



Page 3: Investor Analytics - State-wise investment, transaction type split, age group SIP analysis, monthly volume trends.

Bluestock Fintech — SIP & Market Trends



Page 4: SIP & Market Trends - SIP inflow trend, category inflow heatmap, top categories, folio growth.

7. Recommendations

- **For Retail Investors:** The top-ranked funds on the composite scorecard (Axis Midcap, HDFC Mid-Cap Opportunities, ABSL Frontline Equity) combine strong 3-year returns with healthy Sharpe ratios and should be considered for Moderate-to-High risk portfolios.
- **For Low-Risk Investors:** The recommendation engine identifies ICICI Pru Liquid and Kotak Liquid as suitable Low-risk options with minimal drawdown (under 0.2%) and stable, predictable returns near the risk-free rate.
- **For AMCs - SIP Discipline:** 97.8% of frequent SIP investors show average gaps exceeding 35 days between transactions. AMCs should strengthen auto-debit mandates and SMS/app reminders to improve SIP regularity and reduce churn risk.
- **For AMCs - Geographic Expansion:** While T30 cities dominate current AUM, B30 cities represent an under-penetrated segment. Targeted digital onboarding campaigns in B30 regions could accelerate folio growth.
- **For Portfolio Construction:** All equity funds analysed show HHI below the high-concentration threshold (2500), indicating reasonable sector diversification industry-wide. However, investors holding multiple funds within the same sub-category (e.g. multiple Large Cap funds) should note the 0.85-0.95 NAV correlation, which limits true diversification benefit.

8. Limitations

- Investor transaction data (32,778 records) is synthetically generated, though anchored to real Indian demographic and geographic distributions. Findings on investor behaviour should be interpreted as illustrative rather than exact industry figures.
- The Alpha/Beta computed via OLS regression against Nifty 100 showed very low R-squared values, likely due to date alignment differences between NAV and benchmark series. The pre-supplied `scheme_performance.csv` alpha/beta values were used for the final scorecard as a more reliable source.
- NAV history covers approximately 4.3 years (Jan 2022 - May 2026), which limits the accuracy of 5-year CAGR calculations; 1-year, 3-year and 4-year windows were used instead.
- The dashboard is built in Power BI Desktop and was not published to Power BI Service due to free-tier workspace limitations; the .pbix file and PDF export are provided as deliverables instead.

All data in this project is sourced from publicly available information published by AMFI India, NSE, BSE and open APIs (mfapi.in). This project is for educational purposes only and does not constitute financial advice. Mutual Fund investments are subject to market risks.